Dead Signal

Audio Bible

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Version	1.1
Status	Living — reflects src/audio.js + its callsites in src/DeadSignal.jsx
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Owner	Jharek (audio/design/dev)
Canon source	STORY.md — the Signal motif and tone rules
Companion docs	GDD · Technical Design · PRD

> **Purpose.** The single reference for how *Dead Signal* sounds and why. It defines the audio philosophy, the full palette, the signature **Signal** motif and the rules that govern it, the mix, playback lifecycle, and audio accessibility. All sound is **procedural** (Tone.js) — there are no audio assets to manage; this document *is* the sound design.
>
> **Ground truth.** The engine is src/audio.js; the cue placements are its callsites in src/DeadSignal.jsx. Where this doc gives numbers, they are the shipped values.

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1. Audio philosophy

Audio in *Dead Signal* follows the same restraint as everything else (GDD Pillar 5). Four rules:

1.1 Silence is the bed

There is no looping ambient music or drone — by design. A city that has gone quiet should *sound* quiet. The absence of a soundtrack is the soundtrack. Every sound the game makes is therefore an event, and events in silence carry weight. This is the single most important audio decision.

1.2 Sound is feedback, not decoration

The palette is almost entirely **short UI/feedback one-shots**: a tap, a message blip, a resource sting. They confirm the player's actions and the game's state on a phone-native interface. They are functional first, atmospheric second.

1.3 One sound is allowed to mean something

The **Signal distortion** cue (§5) is the exception that proves the rule. It is the only "atmospheric" sound, it is rare, it is **story-gated**, and it always means the same thing: *the Signal is here.* Because everything else is functional and sparse, this one cue can carry dread.

1.4 The game is fully playable in silence

No critical information is audio-only. Every cue has a textual or visual counterpart (a message appears, a resource number changes, a screen flickers). Audio is **purely additive** — see §8.

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2. The soundscape at a glance

Layer	Sound	Character	Trigger
UI	tapResponse	Soft triangle blip (C5)	Tapping a story choice
UI	tapMenu	Softer sine blip (A5)	Menu / navigation taps
Messaging	blip	Very quiet high sine (E6)	An incoming message arrives
Resources	gain	Rising two-note sting (A4→E5)	Gaining a resource / good outcome
Resources	loss	Low falling sting	Losing a resource /

		(C4)	taking damage
Terminal	terminal("complete")	Consonant resolve chord (Cmaj7)	Completing a run
The Signal	signal	Pink-noise crackle + detuned chirp	**Story beats only** (§5)
Echo recovery *(v2 · PLANNED)*	recovered-voice one-shot	Quieter, degraded Signal sibling	**Recovering an Echo only** (§5.5)

That is the entire soundscape. Seven voices, no loops. **Expansion v2 (build: PLANNED)** adds one story-gated **one-shot** — the recovered-voice Echo cue (§5.5) — and no ambient systems; silence stays the bed (§1.1).

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3. The procedural engine

src/audio.js is a **module-scope Tone.js singleton**. It builds a fixed set of synth nodes **once** on first unlock and retriggers them by name; the game just calls methods like audioEngine.blip().

Key architectural properties:

- **Additive & safe.** Every method is a **no-op until unlocked and while muted**, so game code never needs to guard an audio call. Muting or a failed unlock silences everything without branching.
- **Lazy-loaded.** Tone.js (~380 kB) is import()-ed on the first user gesture (§7), keeping it off the initial bundle.
- **No asset pipeline.** Everything is synthesized. There are no audio files to author, localize, compress, or stream — a deliberate fit for the offline, deterministic product (PRD NFR-1).
- **Signal chain:** voices → master gain → Limiter(-2) → destination, with a Reverb tail branch feeding only the completion chord. (Full mix in §6.)

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4. Sound palette

Exact synth specifications (from src/audio.js). All levels are per-voice dB trims under the master.

4.1 UI — taps

The two tap voices separate *acting on the story* from *navigating menus*.

Voice	Synth	Envelope (a/d/s/r)	Level	Note	Fires at
tapResponse	Triangle	0.001 / 0.12 / 0 /	−9 dB	C5	Story-choice taps

		0.07			(DeadSignal.jsx :2963, :3770, :4495)
tapMenu	Sine	0.002 / 0.15 / 0 / 0.10	−9 dB	A5	Menu/ navigation taps (:4189)

Triangle vs. sine gives the story tap a touch more edge than the menu tap — the player subconsciously hears the difference between "I made a choice" and "I moved through a menu."

4.2 Messaging — the blip

Voice	Synth	Envelope	Level	Note	Fires at
blip	Sine	0.001 / 0.05 / 0 / 0.04	**−24 dB**	E6	An incoming message renders (:2501, :2538)

Deliberately the **quietest** voice in the game (−24 dB) and very short. It reads as a phone's message tick, not an alert — present but never intrusive across a long text conversation.

4.3 Resources — the stings

One triangle "sting" voice, played two ways, so gain and loss are instantly distinguishable by **contour**:

Method	Pattern	Reads as	Fires at
gain()	A4 → E5 (rising, +0.09 s apart)	Relief / pickup	Resource gain, good outcomes (:2546, :3625–3626)
loss()	C4 (low, longer 0.22 s)	Cost / damage	Resource loss, damage (:2546, :2604, :3527, :3627)

Rising interval = good; low single note = bad. No text needed to feel the difference — but the number also changes on the HUD (§8).

4.4 Terminal — the resolve chord

Voice	Synth	Envelope	Level	Chord	Fires at
resolve	PolySynth(Sine)	0.4 / 1.2 / 0.2 / 3.5	−13 dB	**C4·E4·G4·B4** (Cmaj7), 4 s, through reverb	Run completion (terminal("complete"))

The **only** consonant, sustained, reverbed sound in the game. A major-7 chord with a long release — a quiet exhale that marks a completed run. Offline/dead terminals are **silent** (there is no ambient bed to cut, so their silence is the statement).

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5. The Signal — the signature motif

> This is the audio identity of the game. Treat it as canon (STORY.md): players **learn** that *this sound = the Signal.* Protect that association by never spending it cheaply.

5.1 What it is (sound design)

A corrupted-transmission artifact built from three nodes:

- **sigNoise** — a pink-noise burst (NoiseSynth, env 0.004 / 0.2 / 0 / 0.06, **−20 dB**) routed through
- **sigFilter** — a **band-pass at 1400 Hz, Q 1.6** (gives the noise a narrow, radio-static “voice” band), plus
- **sigChirp** — a sine voice (**−22 dB**) playing two detuned notes: **B5** (0.05 s, at +0.02 s) then **F♯5** (0.06 s, at +0.14 s) — a short, falling two-note chirp, like a word half-formed and lost.

The result reads as *something trying to speak through static.* It is quiet by design and routes to master so mute/volume apply.

5.2 The visual coupling (critical)

Every signal() call is paired with a visual flicker (setSigFlicker(true), cleared after ~0.9–1.6 s depending on the beat). The distortion is therefore **audio-visual**, never audio-only — which is both a dread amplifier and the core of its accessibility story (§8). Where the audio crackles, the screen flickers.

5.3 The gating rules (do not break these)

- **Story beats only.** signal() **never** fires on routine actions — not on taps, not on ordinary messages, not on resource changes. If it starts feeling common, it has failed.
- **Reserved association.** It always means the Signal is present/acting. Never use it as a generic “alert.”
- **Sparse.** Across a full playthrough it should fire only at the moments below.

5.4 Where it fires (the complete list)

Every shipped signal() callsite, and what it scores:

#	Beat	Callsite	Why
1	A signal/record143 beat effect —	:1969	The prologue’s first wrongness; the count that can’t be

	incl. the **impossible 143 record**		true
2	A **memory fragment** surfacing	:3813	“a memory surfacing through the Signal”
3	A **Phase 3 truth** uncovered	:3439	The moment a region gives up its answer
4	**Entering the finale**	:3727	The last call begins
5	The **Accept/Refuse** choice resolving	:3743	The decision lands
6	**“i remember you.”** — Ellie’s signature line	:4038	“that sound = Ellie/the Signal,” right up against the words

These map to the canon list in STORY.md §8 (approach/memory/143 record/the call) plus the Phase-3 truth and finale beats. **New signal() calls must clear the §5.3 bar** before being added.

5.5 Echo recovery — the “recovered voice” one-shot — Expansion v2 · build: PLANNED

> Canon: STORY.md §3 (the Echoes) and §4 (“Echoes crack, they don’t confess”). Phase 3 lets the player recover short fragments (2–4 lines) of the Signal’s **142 running minds** — a frozen text thread, a looped voice note, a diary still writing. Recovering an Echo is a **story-gated moment**. This is *narrative-density* work, not a new system: an **additive one-shot**, no new ambient systems, no looping bed (§1.1 holds — the pillar is the ceiling).

Proposed sonic identity — a sibling of the Signal cue, not a new sound. An Echo is a *mind surfaced*, so it should read as the Signal’s own voice, but **quieter and more degraded** — the same family as signal() (§5.1), dialled down and worn thin:

- Reuse the Signal’s architecture (pink-noise burst → the 1400 Hz band-pass “voice” band → a detuned sine chirp) so the ear places it instantly as *the Signal*, then bend it toward **recovery, not presence**: lower the level a few dB **below** signal() (target ~−24 to −26 dB — around the message-blip floor, the quietest register in the game), soften the noise attack, and let the chirp **settle** rather than fall away — a word that *almost* forms instead of one lost. Where signal() is “the Signal is here,” the Echo variant is “a mind surfaced.”
- **One dedicated one-shot**, fired **only** when an Echo is recovered — never on routine actions, never on ordinary node entry, never on Case File opens. It inherits every §5.3 gating rule.
- **Audio-visual, like its parent (§5.2).** Pair it with a flicker — but a **fainter/shorter** one than the full Signal flicker, so recovery reads as distinct from presence and the deaf/HoH beat still lands (§8). Same photosensitivity audit applies.

- The player should **learn to read it** the way they learn signal(): sparse enough that when it arrives, it means *someone is in there.* This resolves the open question in §11 ("a second Signal variant?") **in the narrow, story-gated direction only** — a recovered-voice sibling, not a general "observing vs. acting" split.

Restraint rules for the emotionally heavy Echoes (do not break these). The Echoes are text-forward by canon; the sound design must **not editorialize grief**:

- **The child (Theo) and the regretter (Priya) get NO musical underscoring and NO stinger.** No chord, no swell, no gain/loss sting, no "shock" cue. STORY.md §4: these are the *shortest, hardest* lines in the game precisely because they are unscored. Let the recovered-voice one-shot (or **even nothing** — near-silence is a legitimate choice for these two) carry them, then let the text sit.
- **Never score an Echo for melodrama.** No Echo gets a bespoke sting, a resolve chord, or a stinger for impact. The one-shot marks *that a mind surfaced*, not *how to feel about it* — the sound must never tell the player the beat is sad. Walt's is a chosen mercy, not a tragedy; the mix treats all seven faces the same (a plain recovery), and the **words** do the differentiating.

Kim's absence — E3, the pointed silence — Expansion v2 · build: PLANNED. Canon: STORY.md §3 ("Kim is NOT an Echo") and the Changelog. At the **Signal Core**, the player looks for Kim among the running minds and she isn't there. **The audio for this beat is the Echo/Signal cue conspicuously NOT firing.** Everywhere else a recovered mind surfaces with the recovered-voice one-shot; here — where the player most expects it — there is **pointed silence**, no flicker, no chirp. The engine's default quiet (§1.1) becomes the statement: the one voice the player trusted is the one the Signal can't give back. Do not fill it with a substitute sting or a sad chord; **the missing cue is the cue.**

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6. Mix & headroom

- **Master chain:** master Gain → Limiter(-2) → destination. The limiter at −2 dB catches peaks so overlapping one-shots never clip.
- **Master headroom:** MASTER_DB = -4. Individual voices are trimmed **below** this (−9 to −24 dB), leaving room before the limiter.
- **User controls:** volume (0–1) scales the master; muted forces the master to 0. **Mute always wins** over volume. Both ramp smoothly (applyMasterGain) to avoid clicks.
- **Reverb** exists only as a tail for the completion chord — nothing else is sent to it, so the game stays dry and close (a phone speaker, not a hall).
- **Relative loudness (loudest → quietest):** taps (−9) → loss/resolve (−12/−13) → Signal noise/chirp (−20/−22) → message blip (−24). The things you *do* are the most audible; atmosphere sits under.

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7. Playback lifecycle

Browser autoplay policy blocks sound until a user gesture, so:

- **unlock()** runs on the first pointer/key gesture: dynamically imports Tone.js, awaits Tone.start(), and **confirms the context actually resumed** (retrying resume() once). If it can't confirm a running context, it **bails without marking unlocked**, so the next gesture retries — audio simply arrives one tap later, which is acceptable because it's additive.
- **iOS specifics:** it deliberately does **not** set navigator.audioSession.type = "playback" (that leaks audio to the background and wedges the context). resume() is called on return-to-foreground and on subsequent taps to recover a suspended context.
- **State restore:** mute/volume are read from storage before the first gesture and applied the moment audio unlocks.
- **Diagnostics:** the on-screen ?debug overlay surfaces audioEngine.status() (unlocked/muted/volume/context state/hasNodes).

Because of all this, engineering can call any cue at any time without checking whether audio is ready.

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8. Audio accessibility

Audio is **purely additive** — the game is fully playable muted, which is a first-class design stance (PRD NFR-5). Specifics:

- **Nothing is audio-only.** Message arrivals render text; resource stings mirror a HUD number change; the completion chord accompanies a completion screen.
- **The Signal cue is audio-visual.** Its screen flicker (§5.2) means deaf/HoH players still receive the "the Signal is here" beat. **Action item